

Homework 10: Blood

Paper track. No home lab (blood work stays at practice).

Part A: Read the grid (2 pts each)

Give the ABO/Rh type:

#	Anti-A	Anti-B	Anti-D	Type
1	clump	clump	clump	
2	—	clump	clump	
3	clump	—	—	
4	—	—	clump	
5	—	—	—	

Part B: Facts (1 pt each)

1. Universal red-cell donor? Universal recipient?
2. List the ABO/Rh types from most common to rarest.
3. Type B blood has which antigen and which antibody?

Part C: Genetics (2 pts each)

1. Two type-O parents — what blood types can their children be?
2. A type-AB parent and a type-O parent — what types can the child be? What can it *not* be?
3. A type-A parent (genotype AO) and a type-B parent (genotype BO) — list all possible child types.

Part D: Chemistry (2 pts each)

1. What happens to blood, chemically, as it sits exposed to air? (3 things)
2. What protein carries oxygen in red blood cells?
3. Name two presumptive blood tests and the positive result for each.
4. Is superglue (cyanoacrylate) fuming a test for blood? If not, what is it for?

Part E: Scenario (4 pts)

At a scene, there is the victim's blood (type O+) and a **second** blood sample by the back door that types as **A-**. The four suspects type: O+, A-, B+, AB+.

1. Which suspect does the second sample implicate?
2. Why is the *existence* of a second, non-victim blood sample an important clue?
3. Does the blood type prove that suspect committed the crime? Explain.

Answer Key

Part A: 1. AB+. 2. B+. 3. A-. 4. O+. 5. O-.

Part B:

1. Donor O-; recipient AB+.
2. O+, A+, B+, O-, A-, AB+, B-, AB-.
3. B antigen on the cells; anti-A antibody in the plasma.

Part C:

1. Only type O.
2. Child can be A or B; **cannot** be O and cannot be AB.
3. A, B, AB, or O (all four).

Part D:

1. Oxygen binds the hemoglobin (turns brighter red); platelets activate and it clots; the hemoglobin then oxidizes and the dried stain turns reddish-brown to brown over hours.
2. Hemoglobin.
3. Kastle-Meyer → bright pink; luminol → glows blue in the dark. (Also Hemastix.)
4. No — it develops latent **fingerprints**.

Part E:

1. The **A-** suspect.
2. It suggests the attacker was also injured (e.g., cut on the victim's broken glass or their own weapon) — placing an injured person at the scene.
3. No — blood type is class evidence. A- is fairly uncommon but still shared by millions of people; it narrows the field but doesn't identify one person. DNA from the sample would.